Research defense and next questions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete research-case evidence record.

Separate descriptive results, computational consistency and untested generalization while proposing a controlled follow-up.

Keep one habit: Recall → W1–W3/W5 → independent case → project evidence → feedback and delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.
- Without opening the old lesson, explain one key idea from Presentation and challenge session. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Independent reproduction check. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Writing the methods section. Give an example and a mistake that the idea helps you avoid.

Presentation and challenge session

> **My remembered idea, example and correction**

Independent reproduction check

> **My remembered idea, example and correction**

Writing the methods section

> **My remembered idea, example and correction**

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A research defense distinguishes what was measured, what failed, what was computationally recovered and what remains untested. Main candidate 12/16 = 75%, baseline 11/16 = 68.75%, descriptive gain 6.25 percentage points. The stipulated matching replay supports computational consistency, not a fresh-population or user-benefit claim.

  Separate frozen-model picture gives clear 8/10 = 80%, shaded 4/10 = 40%, a forty-point gap. Source grouping is stated, but object counts and cross-condition pairing are not supplied. Lighting, camera and object differences are plausible explanations, not established causes.

  Propose the same held-out objects under both lights, one camera, fixed scoring, appropriate collection/order controls and justified independent-unit analysis. No such follow-up is run. Changing cameras with lighting entangles explanations again. A limited or negative outcome can complete a sound investigation; retain it and ask a bounded next question instead of manufacturing a positive headline.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

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- Study shows candidate 12/16 versus baseline 11/16; replay matches; no fresh population sample or user study. Show the intermediate steps and give the conclusion with its scope.

Main defense component	Supported statement
Candidate observed rate	12/16 = 75%
Baseline observed rate	11/16 = 68.75%
Descriptive difference	One of sixteen = 6.25 percentage points
Replay matches	Supports recorded computational consistency under the supplied scenario
Fresh population / user benefit	Not studied
Significance / universal usefulness	Not supplied or established

Main defense component	Supported statement
Next work	Resolve a named evidence gap with an explicit new design, not a stronger headline

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

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  Propose the same held-out objects under both lights, one camera, fixed scoring, appropriate collection/order controls and justified independent-unit analysis. No such follow-up is run. Changing cameras with lighting entangles explanations again. A limited or negative outcome can complete a sound investigation; retain it and ask a bounded next question instead of manufacturing a positive headline.

- A report omits this qualification: The **next question** follows **evidence** gaps; it must not be presented as an answer already obtained.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

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  Separate frozen-model picture gives clear 8/10 = 80%, shaded 4/10 = 40%, a forty-point gap. Source grouping is stated, but object counts and cross-condition pairing are not supplied. Lighting, camera and object differences are plausible explanations, not established causes.

  Propose the same held-out objects under both lights, one camera, fixed scoring, appropriate collection/order controls and justified independent-unit analysis. No such follow-up is run. Changing cameras with lighting entangles explanations again. A limited or negative outcome can complete a sound investigation; retain it and ask a bounded next question instead of manufacturing a positive headline.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project candidate 9/12, baseline 10/12; reproduction matches. State result and a useful **next question**.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- For Reproducible AI Research Study, use this week's independent task as a rehearsal for the final **evidence** pack: Project candidate 9/12, baseline 10/12; reproduction matches. State result and a useful **next question**. Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for **Research defense** and **next question**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Research defense** and **next question**s. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project candidate 9/12, baseline 10/12; reproduction matches. State result and a useful **next question**.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

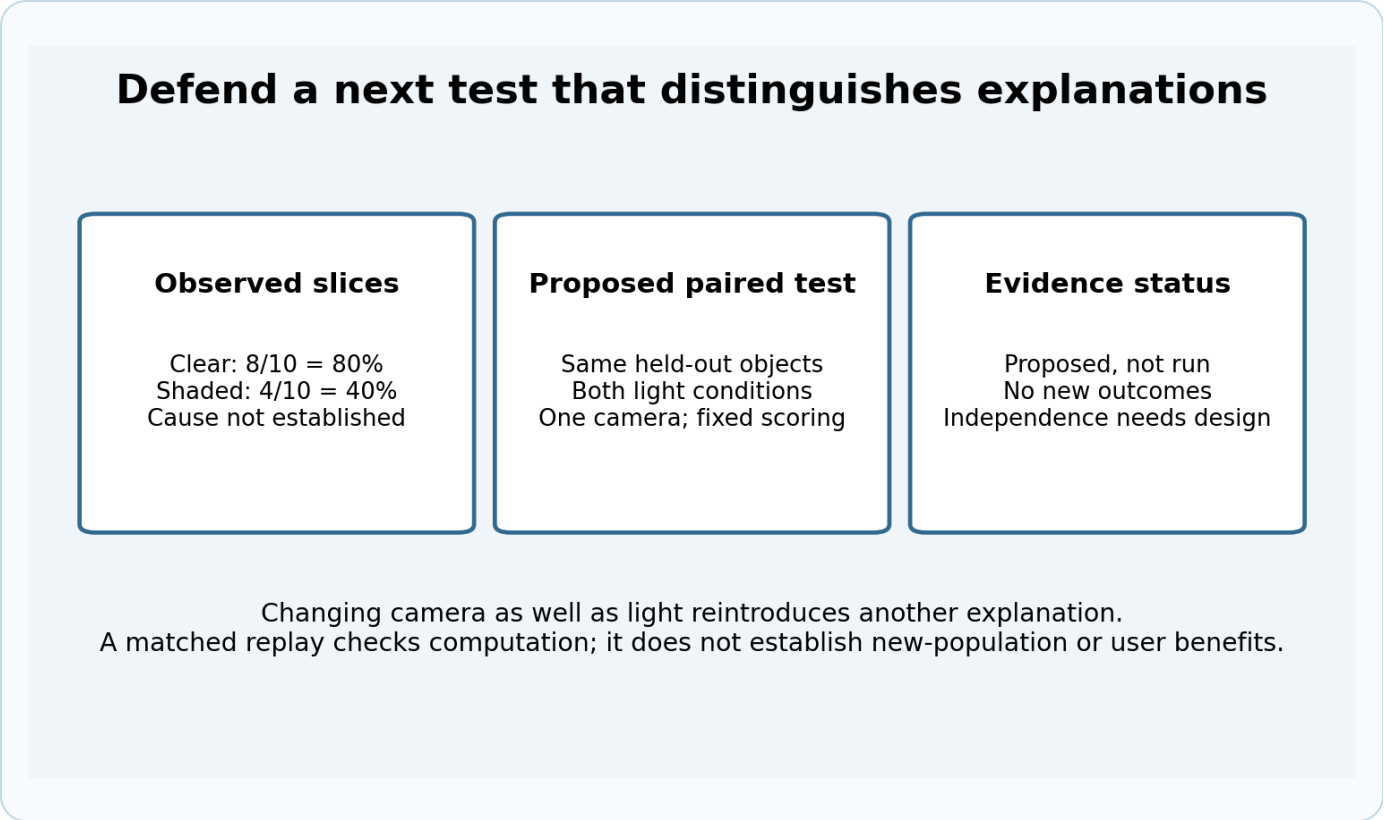
- Defend the next research question. A research defence identifies an observed weakness and a testable follow-up.

- A frozen model is correct on 8/10 clear images and 4/10 shaded images. Source objects are grouped within each condition.

- Compute the clear and shaded rates and identify alternative explanations. How does the same-object, one-camera follow-up help, and why does changing cameras again reintroduce confounding?

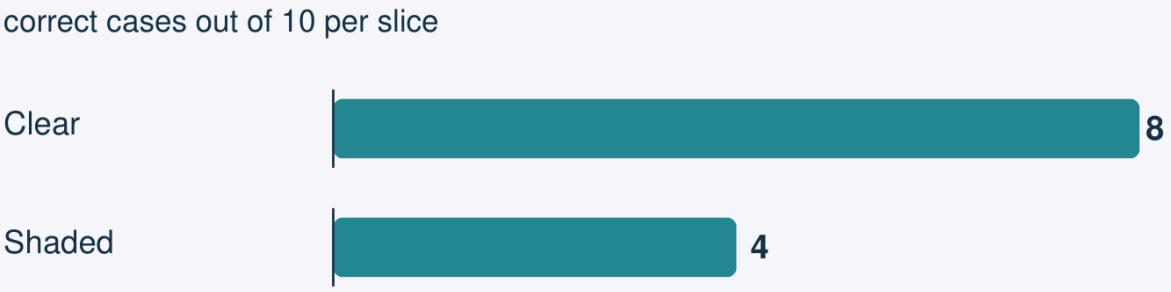
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



AI ACADEMY / AI-7/36

Defence: different condition slices



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/36

Research defense and next questions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. State slice outcomes

Clear is 80%; shaded is 40%.



2. Limit the interpretation

Shade sensitivity is plausible, but camera or object differences may also matter.



3. Design follow-up

Photograph the same held-out objects under both light conditions using one camera and fixed scoring.

CHANGE ONE THING

Change cameras as well as lighting in the follow-up.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ Reference · Project milestone

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Defend the next research question. A research defence identifies an observed weakness and a testable follow-up.
- Which labelled step matches this detail? Photograph the same held-out objects under both light conditions using one camera and fixed scoring.
- 1. Design follow-up
- 2. Limit the interpretation
- 3. State slice outcomes
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Design follow-up Photograph the same held-out objects under both light conditions using one camera and fixed scoring.

A defence should explain results, uncertainty and what would discriminate between explanations next. A controlled paired follow-up is more informative than simply collecting extra easy images.

Changed-case question and optional hint

- Changed-case practice: Change cameras as well as lighting in the follow-up. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. The design again entangles two conditions.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The design again entangles two conditions. A camera difference provides another explanation for any performance gap.

Repair and retain the record

Review Presentation and challenge session (ai-7/35). Goal: Explain the **mechanism** and terminology of presentation and challenge session. Short practice: Answer the camera-B challenge using only the twelve camera-A records.

Why is one displayed camera-B success a demonstration rather than broad **evidence** of camera-B reliability?. Return to this week and explain what you changed.

Review Independent reproduction check (ai-7/33). Goal: Explain the **mechanism** and terminology of independent reproduction check. Short practice: Locate the Q3 scoring disagreement between 5/6 and 4/6.

What should the teams inspect before choosing a score, and what limitation follows if the saved Q3 file is missing?. Return to this week and explain what you changed.

Review Writing the methods section (ai-7/30). Goal: Explain the **mechanism** and terminology of writing the methods section. Short practice: Apply the inclusive four-character rule to oak, pine and willow.

Which label changes under greater-than-four, and what method detail allows another team to reproduce the distinction?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

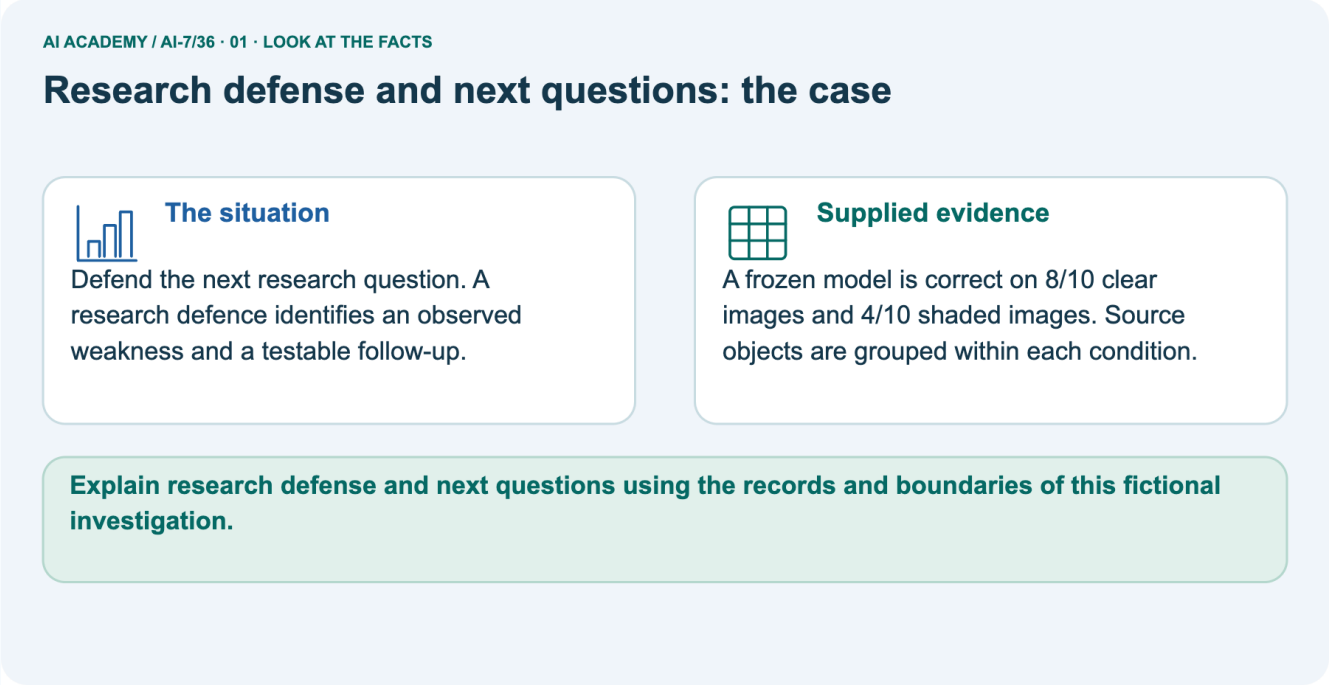
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Research defense and next questions: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

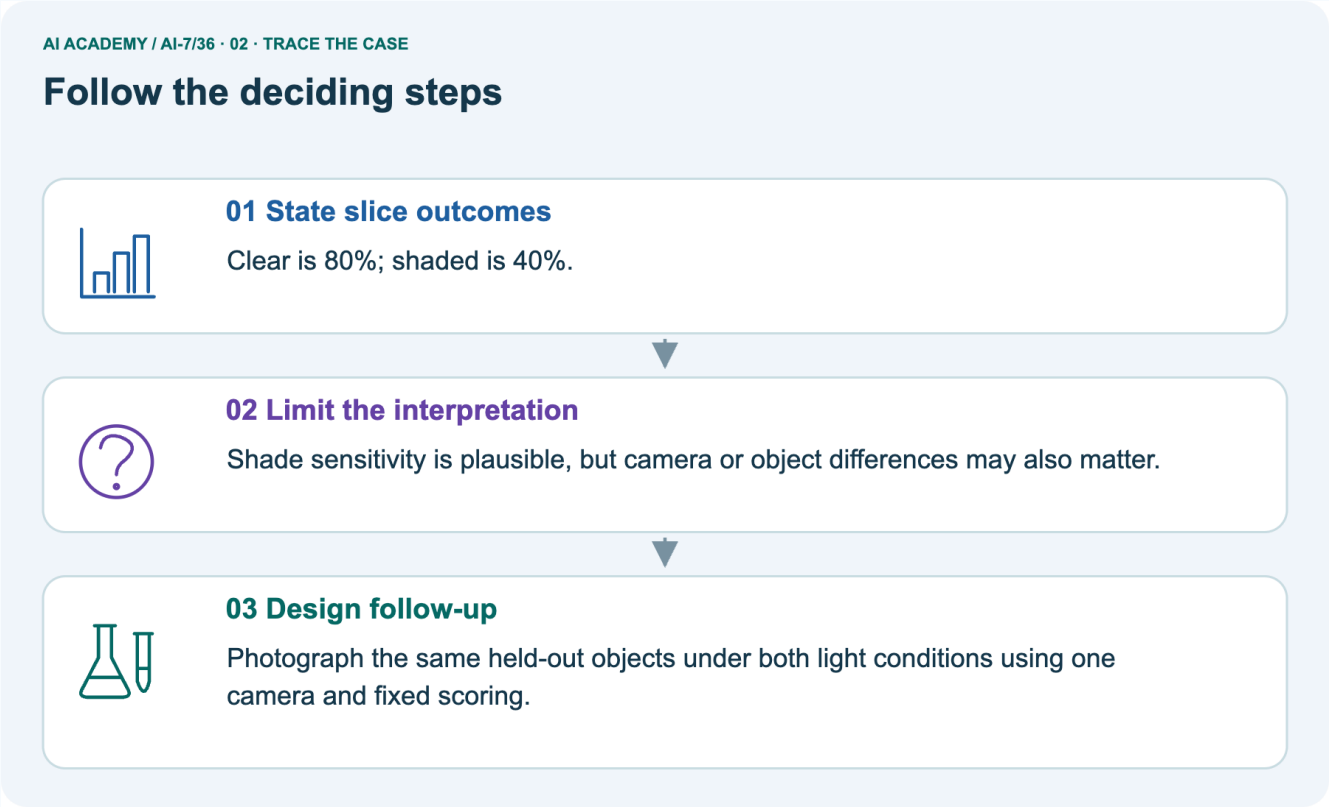
Defend the next research question. A research defence identifies an observed weakness and a testable follow-up. Supplied facts: A frozen model is correct on 8/10 clear images and 4/10 shaded images.

Source objects are grouped within each condition.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



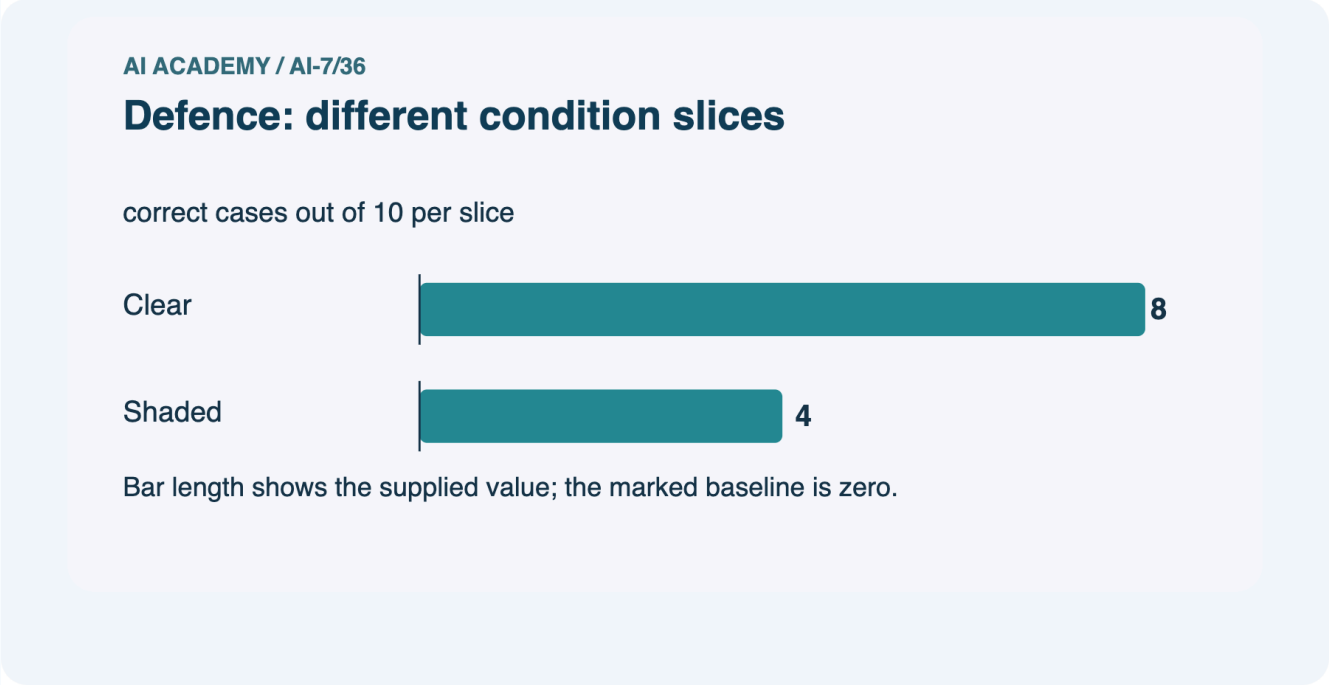
Follow the labelled picture with the matching explanation. Open the vector diagram

State slice outcomes: Clear is **80%**; shaded is 40%. → Limit the interpretation: Shade sensitivity is plausible, but camera or object differences may also matter. → Design follow-up: Photograph the same held-out objects under both light conditions using one camera and fixed scoring.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Defence: different condition slices



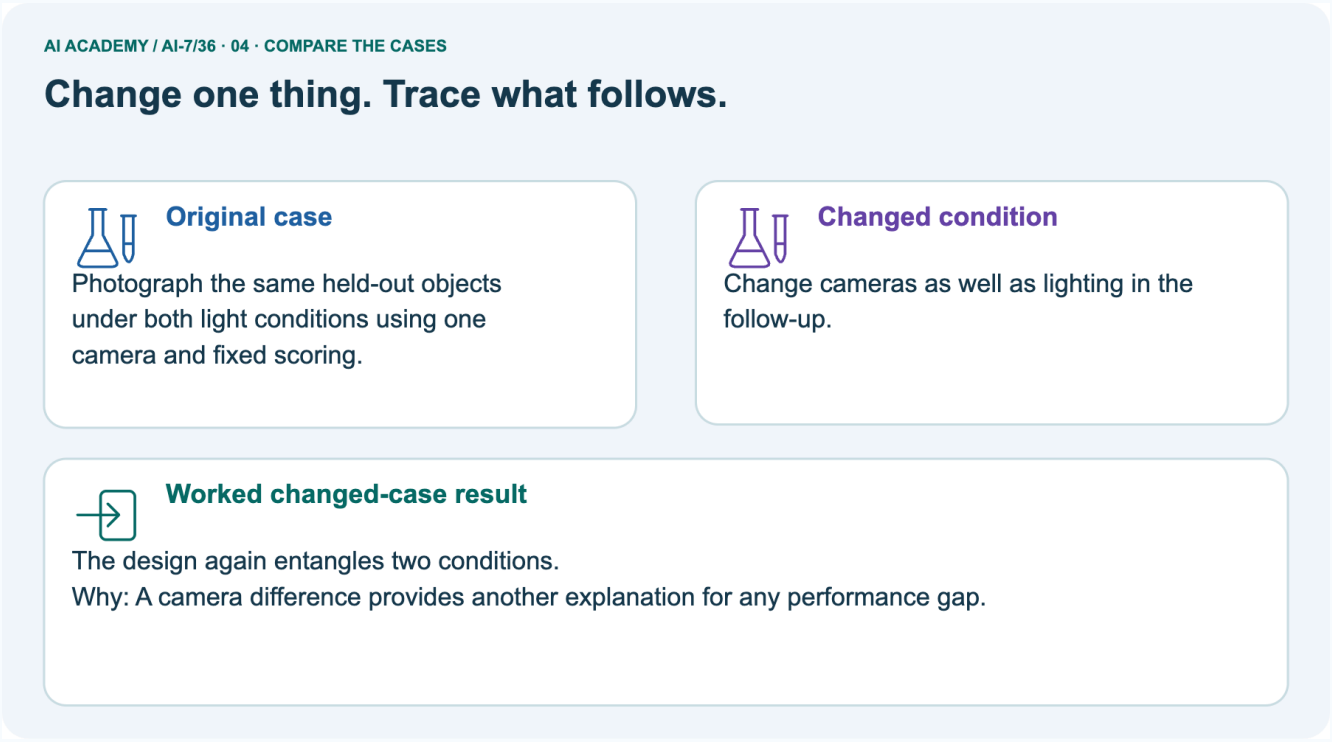
Follow the labelled picture with the matching explanation. Open the vector diagram

Defence: different condition slices. Units: correct cases out of 10 per slice. Clear: 8. Shaded: 4.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

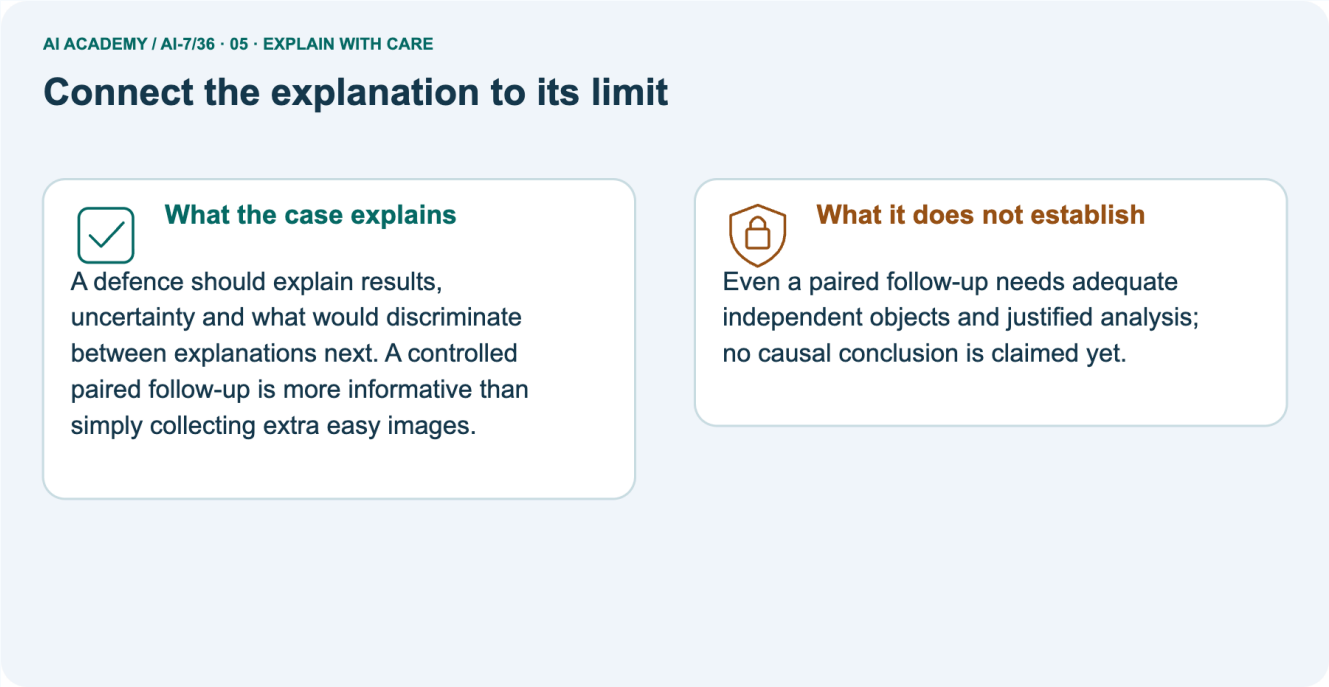
Original case: Photograph the same held-out objects under both light conditions using one camera and fixed scoring. Changed condition: Change cameras as well as lighting in the follow-up.

Expected result: The design again entangles two conditions. Reason: A camera difference provides another explanation for any performance gap.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

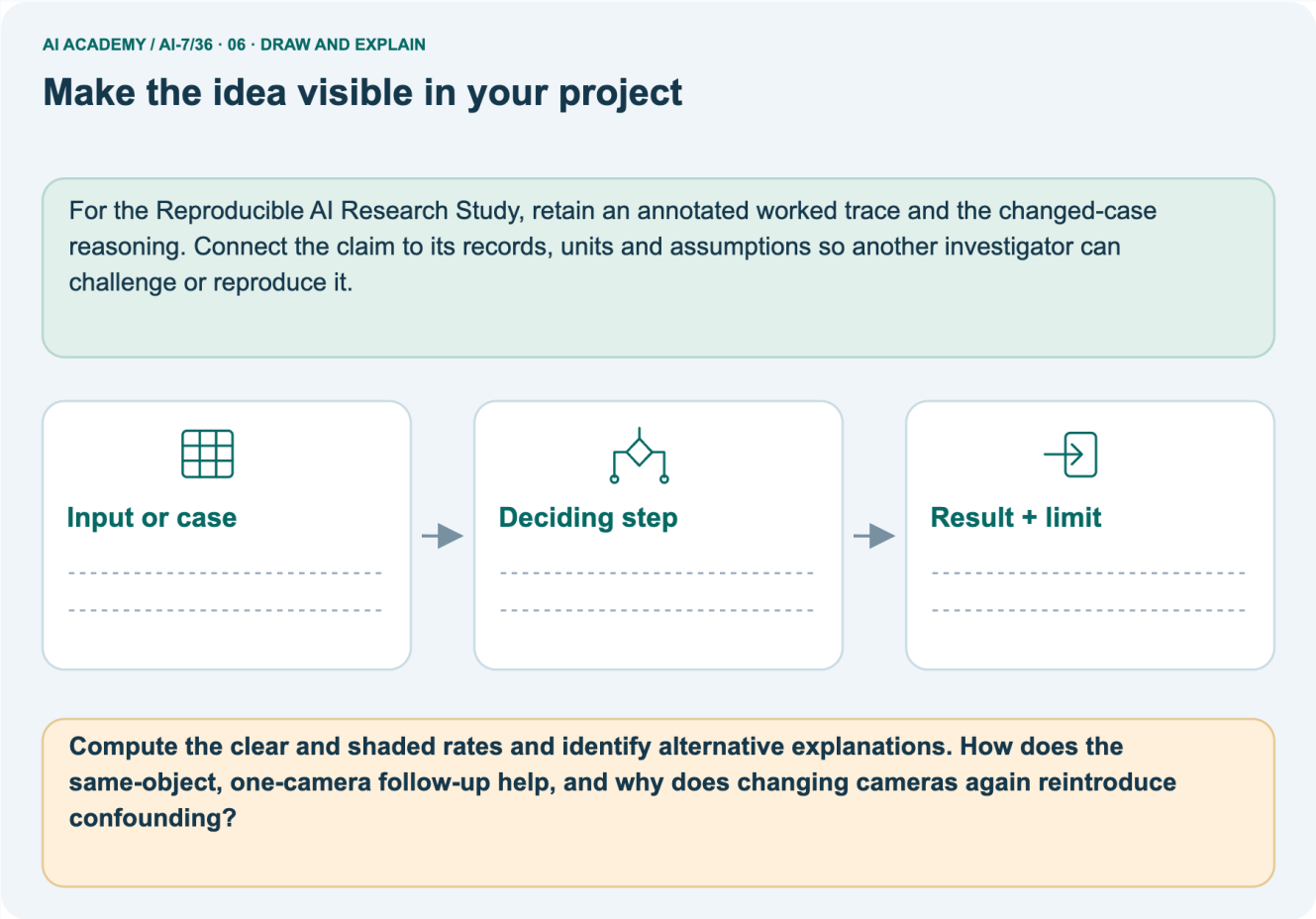
A defence should explain results, uncertainty and what would discriminate between explanations next. A controlled paired follow-up is more informative than simply collecting extra easy images.

Limit: Even a paired follow-up needs adequate independent objects and justified analysis; no causal conclusion is claimed yet.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Compute the clear and shaded rates and identify alternative explanations.

How does the same-object, one-camera follow-up help, and why does changing cameras again reintroduce confounding?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Research defense and next questions: the case

02. Follow the deciding steps

03. Defence: different condition slices

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.